88. Let the number of papers be x. Then, pupil's total score $= 63x$.

$\therefore \quad \dfrac{63x+20+2}{x} = 65 \Rightarrow 2x = 22 \Rightarrow x = 11.$

89. Let the number of boys in the class be x.

Then, $18(x + 20) = 20x + 15 \times 20$

$\Rightarrow \quad 18x + 360 = 20x + 300$

$\Rightarrow \quad 2x = 60 \Rightarrow x = 30.$

90. Let the mean marks of the third group be x. Then,

$\dfrac{25 \times 60 + 50 \times 55 + 25 \times x}{25+50+25} = 58$

$\Rightarrow \quad 1500 + 2750 + 25x = 5800$

$\Rightarrow \quad 25x = 1550 \Rightarrow x = 62.$

91. Average marks in Physics $= \dfrac{60 \times 1 + 75 \times 1 + 85 \times 1 + 95 \times 2}{1+1+1+2}$

$= \dfrac{60+75+85+190}{5} = \dfrac{410}{5} = 82.$

92. Weighted arithmetic mean

$= \dfrac{50 \times 2 + 65 \times 2 + 70 \times 1 + 58 \times 1 + 63 \times 1}{2+2+1+1+1}$

$= \dfrac{100+130+70+58+63}{7} = \dfrac{421}{7} = 60.14 \approx 60.$

93. Let the eighth number be x. Then, sixth number $= (x - 7)$.

Seventh number $= (x - 7) + 4 = (x - 3)$.

So, $\left(2 \times 15\frac{1}{2}\right) + \left(3 \times 21\frac{1}{3}\right) + (x-7) + (x-3) + x = 8 \times 20$

$\Leftrightarrow \quad 31 + 64 + (3x - 10) = 160 \quad \Leftrightarrow \quad 3x = 75 \quad \Leftrightarrow \quad x = 25.$

94. A.M. of 75 numbers = 35. Sum of 75 numbers

$= (75 \times 35) = 2625.$

Total increase $= (75 \times 5) = 375$. Increased sum

$= (2625 + 375) = 3000.$

Increased average $= \dfrac{3000}{75} = 40.$

95. Average of 10 numbers = 7.

Sum of these 10 numbers $= (10 \times 7) = 70$.

$\therefore \quad x_1 + x_2 + \ldots + x_{10} = 70$

$\Rightarrow \quad 12x_1 + 12x_2 + \ldots + 12x_{10} = 840$

$\Rightarrow \quad \dfrac{12x_1 + 12x_2 + \ldots + 12x_{10}}{10} = 84$

$\Rightarrow$ Average of new numbers is 84.

96. $\dfrac{x_1 + x_2 + \ldots + x_{10}}{10} = \overline{x}$

$\Rightarrow \quad x_1 + x_2 + \ldots + x_{10} = 10\overline{x}$

$\Rightarrow \quad \dfrac{110}{100}x_1 + \dfrac{110}{100}x_2 + \ldots + \dfrac{110}{100}x_{10} = \dfrac{110}{100} \times 10\overline{x}$

$\Rightarrow \quad \dfrac{\frac{110}{100}x_1 + \frac{110}{100}x_2 + \ldots + \frac{110}{100}x_{10}}{10} = \dfrac{11}{10}\overline{x}$

$\Rightarrow$ Average is increased by 10%.

97. Correct sum $= (160 \times 35 + 104 - 144)$ cm $= 5560$ cm.

$\therefore$ Actual average height $= \left(\dfrac{5560}{35}\right)$ cm

$= 158.857$ cm ≈ 158.86 cm.

98. Correct sum $= (78.4 \times 25 + 96 - 69) = 1987$.

$\therefore$ Correct mean $= \dfrac{1987}{25} = 79.48.$

99. Correct sum $= (68 \times 20 + 72 + 61 - 48 - 65) = 1380$.

$\therefore$ Correct average $= \left(\dfrac{1380}{20}\right) = 69.$

100. Correct sum $= (40.2 \times 10 - 18 + 33 - 13) = 404$.

$\therefore$ Correct average $= \left(\dfrac{404}{10}\right) = 40.4.$

101. Let there be x pupils in the class.

Total increase in marks $= \left(x \times \dfrac{1}{2}\right) = \dfrac{x}{2}.$

$\therefore \quad \dfrac{x}{2} = (83 - 63) \Rightarrow \dfrac{x}{2} = 20 \Rightarrow x = 40.$

102. Correct sum $= 36 \times 100 + 90 - 40 = 3650$.

Correct average $= \dfrac{3650}{100} = 36.5$. Error $= (36.5 - 36) = 0.5$.

$\therefore$ Error% $= \left(\dfrac{0.5}{36.5} \times 100\right)\% = \dfrac{100}{73}\% = 1.36\%.$

103. Age of the boy sitting in the middle

$= [26 \times 7 - (19 \times 3 + 32 \times 3)] = (182 - 153)$ years

$= 29$ years.

104. Third number $= [306.4 \times 5 - (431 \times 2 + 214.5 \times 2)]$

$= (1532 - 1291) = 241.$

105. Marks obtained by the 11th candidate

$= [(45 \times 22) - (55 \times 10 + 40 \times 11)]$

$= (990 - 990) = 0.$

106. Middle number $= [(10.5 \times 6 + 11.4 \times 6) - 10.9 \times 11]$

$= (131.4 - 119.9) = 11.5.$

107. Temperature on the fourth day $= [(40.2 \times 4 + 41.3 \times 4) - (40.6 \times 7)]$°C $= 41.8$ °C.

108. Total weight of $(A + B + C) = \left(54\dfrac{1}{3} \times 3\right)$ kg $= 163$ kg.

Total weight of $(B + D + E) = (53 \times 3)$ kg $= 159$ kg.
Adding both, we get : $A + 2B + C + D + E = (163 + 159)$ kg $= 322$ kg.
So, to find the average weight of A, B, C, D and E, we ought to know B's weight, which is not given. So, the data is inadequate.

109. Sumit + Krishna + Rishabh $= 43 \times 3 = 129$. ...(*i*)

Sumit + Rishabh + Rohit $= 49 \times 3 = 147$ ...(*ii*)

Subtracting (*i*) from (*ii*), we get: Rohit − Krishna = 18

$\Rightarrow$ Krishna $= 54 - 18 = 36$.

110. $M + T + W = (37 \times 3)$°C $= 111$° C ...(*i*)

$T + W + Th = (34 \times 3)$° C $= 102$° C ...(*ii*)

Subtracting (*ii*) from (*i*), we get:

$M - Th = 9°\text{ C} \Rightarrow M - \dfrac{4}{5}M = 9 \Rightarrow \dfrac{1}{5}M = 9 \Rightarrow M = 45.$

$\therefore$ Temperature on Thursday $= \left(\dfrac{4}{5} \times 45\right)$° C $= 36$° C.